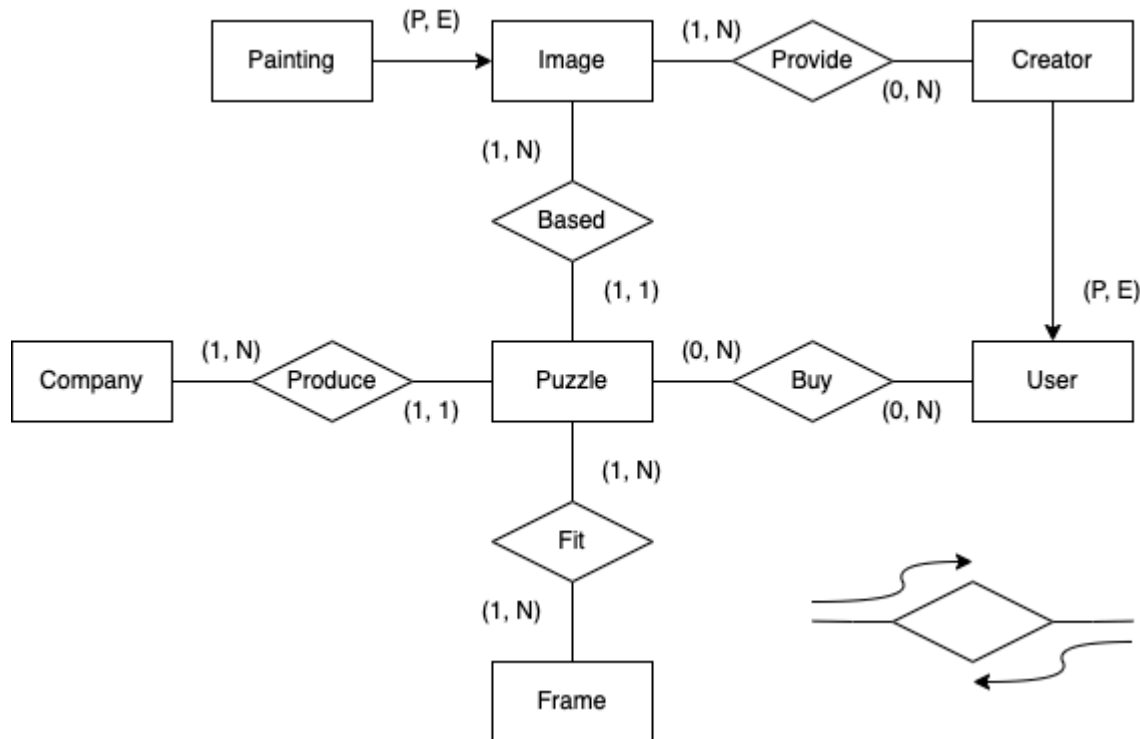


Last Name (Surname, Cognome) _____ First Name _____ Codice Persona _____

Consider the following ER Diagram, representing a data structure for tracking puzzles' production and sales.



The following attributes describe the entities. The primary keys are underlined.

- **Company** - Code, Name, Address, Phone Number, Biography
- **Puzzle** - Puzzle ID, Name, Collection Name (e.g., Museum Collection), Number of Pieces, Width, Height, Description
- **Frame** - Frame ID, Width, Height, Material, Color
- **Image** - Title, Width, Height, Description
 - **Painting** - (...), Artist Name, Artist Surname, Painting Location
- **User** - Personal ID, Name, Surname, Age, Birth Date, City, Country
 - **Creator** - (...), Nickname, Creator Since (e.g., 20/05/2020)

The following attributes describe the relationships.

- **Produce** - Production Date
- **Buy** - Buy Date
- **Provide** - Original Work (e.g., Yes)

N.B. All sizes are in cm.

Exercise 1 - Cassandra (1 PT)

Suppose you expect to record a large number of purchases of puzzles and you decide to store them in a columnar database. Assume you want to run typical queries by searching by the puzzle IDs and by User City or Country. Write the specification of the tables (and other artifacts if necessary) of a Cassandra deployment that stores the required data.

Exercise 2 - Neo4j (5 PT)

Consider the entities User, Creator, Image, Painting, and Puzzle from the ER model.

Suppose you want to store the respective data instances in a graph database.

Sketch a graph model/example describing the nodes, main attributes, and edges. Either show an example graph or a graph with types. (1 PT)

Write a Cypher query to compute the total number of paintings' images uploaded by each creator whose location is "Van Gogh Museum". Consider only the creators who have been creators since a date before 02/01/2024. (2 PT)

Write a Cypher query to compute the total number of puzzles bought by each user who created it (*i.e.*, a puzzle is counted towards a user if they both uploaded the image and bought the resulting puzzle), considering only users who bought at least 50 puzzles, whose number of pieces is greater than 1'000. (2 PT)

Exercise 3 - Elasticsearch (3 PT)

Consider the Puzzle entity. Provide the complete mapping of the index for the considered entity (i.e., field name, field type, the structure of the mapping, etc.) (1 PT)

PUT /

Write a query to compute the total number of puzzles for each collection name. (2 PT)

Exercise 4 - SparkQL (3 PT)

Consider the following Frame and Puzzle tables.

<u>Frame ID</u> (PK)	...	...
X		
Y		

<i>Frame ID</i> (FK)	<u>Puzzle ID</u> (PK)	...
[X, Y, ...]		

Write a SparkQL query to return the frames that perfectly fit the puzzle they are fit for. (1.5 PT)

Write a SparkQL query to count the total number of (distinct) frames with a width and height greater than 100 and 70 cm, respectively, that are associated with a puzzle from the "Landscape Collection". (1.5 PT)

Exercise 5 - MongoDB (4.5 PT)

Consider the entities Puzzle, Image, Frame, and Company. How many collections would you define within a documental database (clearly state it)?

How would you implement the relations between the concepts? Provide a simple documental representation. (1 PT)

Write a query to compute the total number of puzzles produced by each company, considering only puzzles produced after 30/04/1995 and with more than 500 pieces, and returning only the companies that produced more than 100 puzzles (1.5 PT)

Write a query to compute the total number of (distinct) frames associated with each puzzle based on a picture from "Leonardo da Vinci" and whose location is "Louvre", returning only the puzzles with more than 10 frames. (2 PT)